

Project Title

Where Should You Work? A Data-Driven Location Choice Tool

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Public app URL

https://zhangcx.shinyapps.io/Project_Prototype/

What the app currently does

The current prototype allows users to compare economic conditions across several U.S. states using an interactive Shiny app. Users can select different comparison metrics, including real purchasing power, rent burden, and unemployment rates, and adjust an expected salary level using a slider.

Based on the selected salary level, the app calculates cost-of-living-adjusted salary measures and estimated housing burdens across states. The prototype currently includes comparisons for five states and provides both graphical and table-based outputs.

Initial visualization/table/output

The app currently produces interactive bar charts that compare states based on real purchasing power, estimated rent burden, or unemployment rates. These visualizations are designed to help users understand how differences in prices and housing costs affect the value of income across locations.

The app also includes an interactive data table displaying the underlying indicators used in the calculations, including average salary, median rent, unemployment rate, price index, and user-adjusted affordability measures.

What remains unfinished

Several planned features still need to be completed before the final submission. The current prototype uses manually entered sample data, and the next stage of the project will involve integrating real API-based datasets from sources such as the Bureau of Labor Statistics (BLS), Bureau of Economic Analysis (BEA), and the Census Bureau.

In addition, the app currently includes only five states, so expanding the geographic coverage is an important next step. Future improvements may also include occupation-specific filtering, state rankings, improved visual design, and potentially map-based visualizations to make comparisons more intuitive.

Known problems or limitations

The current version uses simplified manually entered data rather than fully automated real-time API integration. Some variables and table labels also need to be clarified to improve readability for users. For example, certain measures such as rent burden and real salary currently include both fixed state-level values and user-adjusted values, which may not yet be fully clear from the table labels alone.

Another limitation is that the current prototype includes only a small number of states, so the comparisons are still limited in scope. The app also does not yet include additional factors such as taxes, occupation-specific salaries, or broader cost-of-living measures.